Narcissus Keyed

Seventeen keyed copies of one convex space-filler force a stacking that never repeats



DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com · ORCID 0009-0001-2620-2036
Paper #97 · 6 November 2026 · DICTO DOI: 10.5281/zenodo.24005567 · Licence CC BY-SA 4.0 · Data, scripts and checks: https://github.com/DICTOR-Master/kaleidohedra (branch research, study 2026-11-05-one-piece-any-stacking)
DICTO set the aim (one piece, any stacking: a single piece whose keys force a stacking that never repeats), found and named Narcissus, proposed that it might be halved and turned, and chose to carry the layer key on Narcissus. Computation, proofs and drafting with Claude (Anthropic).

Stack layers of bricks and you may choose, at each layer, how the next one sits. If every choice is allowed, nothing stops you from repeating the same pattern for ever. Here one solid, DICTO's Narcissus, fills space in layers with exactly two ways to place each new layer: straight or turned. We cut small keys into its faces, in seventeen versions, so that the keys still let the layers fill space but no longer let the sequence of straight and turned repeat. The keys carry a little arithmetic: each layer multiplies a hidden number by 2 or by 2/3, and since no product of 2s and 2/3s equals 1, the number never returns, and neither does the stacking.

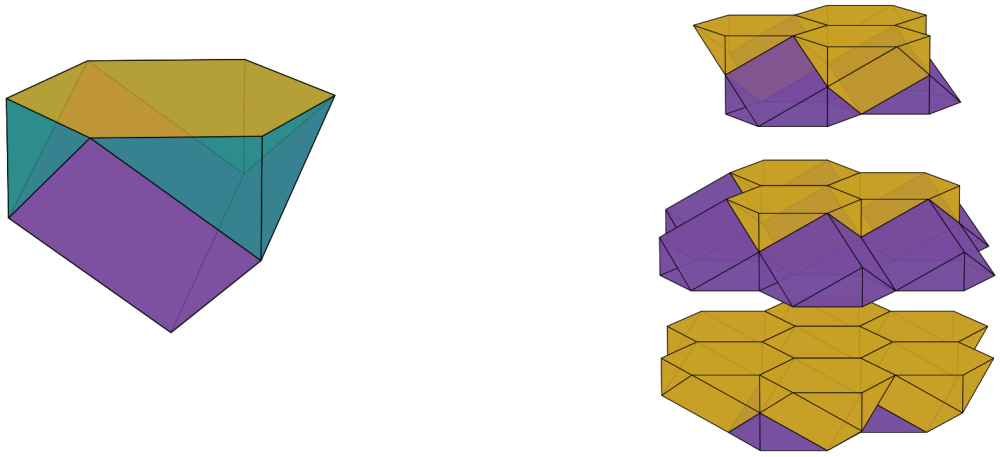
Abstract. Narcissus, the ten-cornered piece of the HexOct h-tile, is its own mirror image and fills space alone. **Theorem 1:** copies of Narcissus fill space in layers, and each layer may be put on the one below in exactly two ways, straight or turned by 60°, freely and independently, so every sequence of straight and turned joins is a tiling. **Theorem 2:** there is a set of seventeen keyed copies of Narcissus (fourteen carrying Kari's aperiodic Wang tiles, three relays) such that keyed tilings exist and in every one the sequence of joins is not periodic; no keyed tiling has a period with a vertical component. The proof is Kari's density argument, carried on the real faces: the join type is the row's multiplier. We check both theorems on the actual solids (sampling, and a SAT search over all six-layer sequences). Narcissus is Goldberg's space-filler type 10-XII.

1. Narcissus and its two joins

At the end of the HexOct walk (#75, $\lambda \geq 3/8$) each h-layer tile is two copies of one piece, Narcissus (DICTO, 4 November 2026): a regular hexagon of side $1/\sqrt{3}$ on top, three corners half a layer lower under alternate hexagon corners, and an apex one layer down below the hexagon's centre (unit ball spacing). It has ten faces (one hexagon, three quadrilaterals, six triangles), ten corners and eighteen edges, volume $\sqrt{2}/4$, dihedral angles 90°, 120° and 135° only, and three mirror planes. Its combinatorial type, formula 1036-64 with the hexagon ringed by six triangles, is Goldberg's space-filler 10-XII [Go82]. Two copies joined on their hexagons make the twisted cube of #86 when put on straight and Covenant, a twelve-parallelogram parallelohedron, when turned by 60°, a four-zone parallelohedron of Fedorov's kind [Fe85, Gr10] (Narcissus and Covenant, HexOct chapter, draft). Goldberg builds 10-XII from two opposite-hand heptahedra 7-XXIX [Go78].

(a) Narcissus

(b) three layers, pulled apart



DICTO

Figure 1. (a) Narcissus: hexagon (gold), three quadrilaterals (purple), six vertical triangles (teal). In the keyed copies the hexagon carries the input digit and three bumps, one quadrilateral the output digit, two opposite triangles the carries. (b) Three layers of Narcissus pulled apart: down-pointing copies gold, up-pointing purple. The lower join is straight, the upper turned by 60°.

2. Narcissus alone, any sequence of joins

Theorem 1. Let a layer be the down-pointing copies of Narcissus on one triangular lattice of sites and the up-pointing copies on the next, between two parallel planes. Each layer fills the space between its planes, and both planes are tiled by its hexagons. A layer can be put on the one below in exactly two ways that keep face-to-face contact: its mirror image in the shared plane (straight) or that mirror image turned by 60° about a hexagon centre (turned). Every sequence of straight and turned joins fills space.

Proof. In the h-tiling of HexOct (#75) the layer is a slab of the tiling, so it fills the space between its planes; its down-pointing copies end in hexagons on the top plane and its up-pointing copies in hexagons on the bottom plane, each a honeycomb of regular hexagons. A layer placed on the top plane must carry the same honeycomb on its lower plane; the placements that do so are the mirror image composed with a symmetry of the honeycomb. The honeycomb's symmetries that are not symmetries of the layer form a single class, represented by the 60° turn about a hexagon centre (it moves the three corners of each hexagon that lie over the mid corners onto the other three). So there are exactly two placements, and since each layer fills a slab between planes, any sequence of slabs fills space. □

The free choice per layer is the law of #92 (Aristotle's Pyramid in Layers) and of the close-packed stackings [Gr94], here for one piece. All straight gives the h-tiling of HexOct; all turned gives Covenant pairs (the fcc lattice of Covenants). Checked on the solids: for ten sequences of five or six joins, including both pure ones and random mixtures, every one of 20 000 random points lies in exactly one copy. Each copy touches exactly ten neighbours face to face: the copy across the hexagon, the three opposite-pointing copies of its layer on its quadrilaterals, and the six same-pointing copies of its layer on its vertical triangles.

3. The keys

Kari's fourteen Wang tiles [Ka96] are tuples (q, n, s, w, e): a multiplier q ∈ {2, 2/3}, digits n and s, carries w (in) and e (out) with q·n + w = s + e. Kari's digits are Beatty differences ⌊kx⌋ − ⌊(k−1)x⌋, the balanced (Sturmian) words of slope x [Lo02]; the same multiplication later served the undecidability of tiling [Ka07]. A row of tiles multiplies the average of its digits by q: by 2 for averages in [1/2, 1], by 2/3 for averages in [1, 2]. We put one row of tiles in each layer, on the down-pointing copies, and use the up-pointing copies to pass digits to the layer below.

Face	Tile piece (one per Kari tile, 14)	Relay (digit d = 0, 1, 2; 3)
Hexagon	receives the input digit n; three bumps at the corners not over the mid corners (q = 2) or at those over them (q = 2/3)	gives the digit d; three hollows at the corners over its mid corners
Quadrilaterals	one gives the output digit s; two carry the key "tile"	one receives the digit d; two carry the key "relay"
Triangles	carry out e (one), carry in w (the opposite one), four plain	six plain (relay)

Keys mate as follows: digit to equal digit; carry to equal carry of the same multiplier; "tile" to "relay"; plain to plain. A bump fits a hollow. Under a straight join the corners of a hexagon over its mid corners meet the same kind of corners; a turn by 60° exchanges the two kinds. So a ×2 tile piece fits its relay only across a turned join and a ×2/3 tile piece only across a straight one: **the join type is the row's multiplier**.

4. The stacking never repeats

Theorem 2. The seventeen keyed copies of Narcissus tile space, and in every tiling the sequence of straight and turned joins is not periodic. In particular no tiling by them is invariant under a translation with a vertical component.

Proof. *Roles.* The quadrilaterals of a layer join down-pointing to up-pointing copies, in a connected honeycomb pattern; "tile" mates only "relay" and the digit keys mate only each other, so in each layer all copies of one kind are tile pieces and all of the other kind relays, and the hexagons (which mate a tile piece only with a relay) carry this through every layer. We write the proof for tile pieces pointing down; the other case is the same read upwards. *Rows.* In a layer the carries run through opposite triangles, so the tile pieces form parallel straight chains with one direction per layer; each piece's multiplier is fixed by its hexagon, and the join above the layer is the same for all its pieces, so one layer has one multiplier q. *Density.* Over a large patch of a layer, the sum of q·n − s over its tile pieces equals the carries leaving the patch minus those entering, which is bounded by a constant times the patch's perimeter. So the averages obey $\bar{s} = q \cdot \bar{n}$. Each tile piece sends s through one quadrilateral to a relay, which hands it through its hexagon to the tile piece below, a one-to-one shift of the lattice; so the average input of the layer below is $\bar{s}$. Writing x_k for the average input digit of layer k, $x_{k-1} = q_k \cdot x_k$, with $q_k = 2$ if the join above layer k is turned and 2/3 if straight. Kari's digits keep x_k between 1/2 and 2 (×2 layers give outputs 1 or 2; ×2/3 layers take inputs 1 or 2). *No repetition.* If the joins repeated with period P, then $x_k = \rho^m \cdot x_{k+mP}$ for every m with $\rho = 2^a(2/3)^b \neq 1$ ($a + b = P$, and $2^{a+b} = 3^b$ only for $P = 0$); for m of one sign x would leave [1/2, 2]. *Existence.* Kari's tilings of the plane by his fourteen tiles give a layer for each row. In a layer above a turned join every copy is turned by 60°, while a single copy can turn only by 120° about its axis; so the chains in such layers run in the opposite

direction and the row is read right to left. Kari's tiles also tile such reversed rows: with $g(i) = q\lfloor ix \rfloor - \lfloor iqx \rfloor$, the carries $K - g(i)$ with $K = -1$ ($q = 2$) and $K = 1/3$ ($q = 2/3$) use the same fourteen tiles. □

5. Checks on the solids

We placed the seventeen keyed copies on the actual layers by proper rigid motions of one reference copy (three per site), read the keys off the real faces and the real face-to-face contacts, (rigid placements by Kabsch's method [Ks76]; faces from convex hulls [BDH96]; contact areas by polygon clipping [SH74]) and asked a SAT solver (CaDiCaL [Bi20] through PySAT [IMM18]) whether a keyed tiling of a finite block exists for each sequence of joins. For blocks ten units wide, two rows deep and seven layers high, 48 of the 64 sequences of six joins admit a keyed tiling. Each of the sixteen that fail contains three turned joins in a row, which Kari's arithmetic forbids: three ×2 layers would take an average from [1/2, 1] beyond 2. Four sequences that contain three turned joins still pass, where the run reaches the open top layer. The reversed-row identity above was checked exactly (rational arithmetic) on 300 columns for every sampled average.

6. What is known and what is new

Known: Wang's tiles and his question [Wa61], Berger's first aperiodic set and the undecidability of tiling [Be66], Robinson's six tiles [Ro71], the survey of Grünbaum and Shephard [GS87]; aperiodic Wang tiles and Kari's multiplication method [Ka96], Culik's variant [Cu96], the minimum of eleven Wang tiles [JR21], aperiodic Wang cubes [CK95]; weak and strong aperiodicity [GS98]; the Schmitt–Conway–Danzer biprism, which forces layers turned by an irrational angle [Da95]; the Socolar–Taylor tile and its three-dimensional form, whose non-periodic layers stack periodically [ST11, ST12]; the hat, an aperiodic monotile of the plane [SMKG24]; a translational aperiodic tile in very high dimension, disproving the periodic tiling conjecture [GT24]; Goldberg's space-filling decahedra, including the type of Narcissus [Go82]; and, as preprints, a strongly aperiodic monotile in three dimensions, a non-convex seven-cube chair with face features [Ts26], with matching rules [Fl26]. New within the sources checked: that Narcissus alone fills space for every sequence of straight and turned joins (Theorem 1), and that keying one convex space-filler so that its own join type is Kari's multiplier forces the stacking itself never to repeat (Theorem 2). The aperiodicity here is in the stacking direction only: a keyed tiling may repeat within its layers. A Culik-type set of twelve tiles (×3 and ×1/2) was tried and rejected: it allows an all-zero layer, hence a periodic tiling. Fewer keyed copies, ideally one, remain open.

Sources checked: Goldberg, On the space-filling decahedra (free copy, UPC); Kari 1996 and Culik 1996 (through their tile rules, reproduced and checked here); Jeandel–Rao; the Radboud thesis on Wang cubes; Wikipedia pages on aperiodic sets of prototiles and the Socolar–Taylor tile; arXiv abstracts of Tsiokos and Flicker. Web searches: "aperiodic set of polyhedra forcing non-periodic stacking layers Kari Wang tiles three dimensions keyed copies single shape"; "Socolar Taylor three-dimensional realization aperiodic monotile layers stacking weakly aperiodic Schmitt Conway Danzer". No construction coupling a polyhedron's join type to a Wang multiplier was found.

References

[BDH96] C. B. Barber, D. P. Dobkin, H. Huhdanpaa, The Quickhull algorithm for convex hulls, ACM Transactions on Mathematical Software 22 (1996) 469–483. — [Be66] R. Berger, The undecidability of the domino problem, Memoirs of the American Mathematical Society 66 (1966). — [Bi20] A. Biere, K. Fazekas, M. Fleury, M. Heisinger, CaDiCaL, Kissat, Paracooba, Plingeling and Treengeling entering the SAT Competition 2020, Proc. SAT Competition 2020, 51–53. — [CK95] K. Culik II, J. Kari, An aperiodic set of Wang cubes, J. Universal Computer Science 1 (1995) 675–686. — [Cu96] K. Culik II, An aperiodic set of 13 Wang tiles, Discrete Mathematics 160 (1996) 245–251. — [Da95] L. Danzer, A family of 3D-spacefillers not permitting any periodic or quasiperiodic tiling, in Aperiodic '94 (World Scientific, 1995). — [Fe85] E. S. Fedorov, Начала учения о фигурах (Elements of the Study of Figures), St Petersburg (1885). — [Fl26] F. Flicker, Matching rules for a three-dimensional strongly aperiodic monotile, arXiv:2609.23783 (2026, preprint). — [Go78] M. Goldberg, On the space-filling heptahedra, Geometriae Dedicata 7 (1978) 175–184. — [Go82] M. Goldberg, On the space-filling decahedra, Structural Topology 7 (1982) 39–44. — [Gr10] B. Grünbaum, The Bilinski dodecahedron and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra, Mathematical Intelligencer 32 (2010) 5–15. — [Gr94] B. Grünbaum, Uniform tilings of 3-space, Geombinatorics 4 (1994) 49–56. — [GS87] B. Grünbaum, G. C. Shephard, Tilings and Patterns, W. H. Freeman (1987). — [GS98] C. Goodman-Strauss, Matching rules and substitution tilings, Annals of Mathematics 147 (1998) 181–223. — [GT24] R. Greenfeld, T. Tao, A counterexample to the periodic tiling conjecture, Annals of Mathematics 200 (2024) 301–363. — [IMM18] A. Ignatiev, A. Morgado, J. Marques-Silva, PySAT: a Python toolkit for prototyping with SAT oracles, SAT 2018, LNCS 10929, 428–437. — [JR21] E. Jeandel, M. Rao, An aperiodic set of 11 Wang tiles, Advances in Combinatorics (2021). — [Ka07] J. Kari, The tiling problem revisited, Machines, Computations and Universality 2007, LNCS 4664, 72–79. — [Ka96] J. Kari, A small aperiodic set of Wang tiles, Discrete Mathematics 160 (1996) 259–264. — [Ks76] W. Kabsch, A solution for the best rotation to relate two sets of vectors, Acta Crystallographica A32 (1976) 922–923. — [Lo02] M. Lothaire, Algebraic Combinatorics on Words, ch. 2 Sturmian words, Cambridge University Press (2002). — [Ro71] R. M. Robinson, Undecidability and nonperiodicity for tilings of the plane, Inventiones Mathematicae 12 (1971) 177–209. — [SH74] I. E. Sutherland, G. W. Hodgman, Reentrant polygon clipping, Communications of the ACM 17 (1974) 32–42. — [SMKG24] D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss, An aperiodic monotile, Combinatorial Theory 4 (2024). — [ST11] J. E. S. Socolar, J. M. Taylor, An aperiodic hexagonal tile, J. Combinatorial Theory A 118 (2011) 2207–2231. — [ST12] J. E. S. Socolar, J. M. Taylor, Forcing nonperiodicity with a single tile, Mathematical Intelligencer 34 (2012) 18–28. — [Ts26] I. Tsiokos, A strongly aperiodic monotile in three dimensions, arXiv:2609.19214 (2026, preprint). — [Wa61] H. Wang, Proving theorems by pattern recognition II, Bell System Technical Journal 40 (1961) 1–41. — DICTO, HexOct (#75), The HexOct Family (#86), Narcissus and Covenant (HexOct chapter, draft).